

Deepika N

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ABSTRACT

Many dog health issues go unnoticed or are detected late because most systems rely on only one type of input. In this project, the idea is to fix that by using a setup that looks at both the dog's photo and the symptoms the owner explains. The image part is checked through deep-learning methods, while the symptoms are handled with a simpler machine-learning approach. When both of these are combined, the system gives results that are earlier and usually more trustworthy for dogs of different sizes and breeds. The whole point is to make things a bit easier for vets and even for pet owners, so they can catch problems sooner instead of waiting until the dog gets worse.

The project is centred on creating a straightforward system that collects a dog's health details, cleans and organizes the data, and then trains models that learn from both the photos and the symptoms provided. Since different breeds and sizes of dogs show health issues differently, the system tries to adjust its predictions based on that. It also gives owners an easy page where they can upload a picture of their dog and mention any signs they've noticed, and it quickly shows what the issue might be. Later on, this work could grow into a mobile app so that owners can keep track of their dog's health in real time. Overall, the project sets the groundwork for a practical, AI-based health monitoring system that understands breed differences and makes checking a dog's condition much easier.

CHAPTER 1

INTRODUCTION

Our project mainly focuses on creating an AI-based Dog Health Monitoring System that can look at both a dog's photo and the symptoms shared by the owner to detect diseases more accurately. Instead of depending only on images or only on text like many current systems do, we use a dual-input model so the assessment becomes more complete for small, medium, and large breeds. Once the visual data is paired with the symptoms, the system can understand the dog's condition much faster and give more useful guidance to vets and pet owners.

The whole idea is to use multimodal AI in a practical way so that dog health monitoring becomes more reliable and actually helps in improving everyday pet care.

PROBLEM STATEMENT

With more and more people depending on their pets for companionship and emotional support, the health management of pets is a growing concern for millions of pet owners. Highly expressive animals, dogs usually manifest subtle changes in behavior, appetite, physical appearance, or energy levels as early signs of some sort of discomfort or illness. Still, the majority of such indicators are not that easy for non-professionals to interpret correctly. Unlike humans, they cannot express feelings of pain or symptoms; it is up to the dog owners to second-guess what was causing the problem based on limited knowledge or information from the internet, which often leads to inaccurate assumptions or delayed medical attention.

Traditional veterinary care often requires the owners to physically see a clinic, even for minor complaints such as sudden vomiting, skin irritation, abnormal stool, or a change in activity level. Although it is crucial to visit a veterinarian in serious cases, this may be very expensive for every small symptom, especially if a family owns more than one pet. Besides, there are also other limitations that some pet owners may face: busy schedules, lack of transportation, or limited availability of veterinary facilities within the place of their residence. Such situations usually lead to a delay in seeking medical consultation until the symptoms become intense, which compromises the possibility of early diagnosis and timely treatment.

Most of the digital pet-health tools and mobile applications available today address only one aspect of the problem. For instance, some apps provide image-based diagnosis of skin diseases, while others use rule-based systems to answer text-based symptom queries. On the other hand, symptom-only systems fail to identify diseases that are supported by important visual cues like lesions, fur loss, rashes, or discoloration. Lack of a system considering both forms of evidence together severely limits the reliability and accuracy of existing solutions.

Another big limitation that exists today is the absence of breed-informed diagnostic tools. Symptoms vary between breeds and breed sizes (small, medium, large), as does their

susceptibility to diseases and their appearance. Certain skin conditions will look very different in short-haired versus long-haired dogs, and appetite loss on a large breed can be very different in implication from the same condition on a small breed. Without the information provided by breed-specific factors, single-input systems give inconsistent and sometimes misleading results.

The lack of multimodal datasets further complicates the development of accurate health monitoring tools. Most available dog-health datasets include symptom descriptions or images, but rarely both together. Because of this, existing AI models are not learning the natural relationship between what owners observe-symptoms-and what the dog visually presents-images. This sets up a significant gap in research findings and points to the need for a unified, multimodal approach.

The above challenges can be addressed with a dual-model AI-based Dog Health Monitoring System that can take both images and symptom descriptions. Dual input processing applied in the proposed model relies on deep learning image analysis for the visible abnormalities while, at the same time, machine learning and NLP techniques can interpret the textual symptom descriptions that the owner provides. These two forms of evidence are then combined by the multi-modal AI to generate a more accurate and holistic health assessment. It will not replace veterinarians, but rather act like a preliminary screening tool to aid the decisions that pet owners make. our system aims to help users understand whether a symptom is common and can be monitored at home, or it indicates wheather it is a serious condition that requires veterinary care. It helps pet owners with action, insights, reduces unnecessary clinic visits, and helps in early detection of diseases. Our work responds to this need by offering an intelligent, breed-aware, user-friendly system able to provide trustworthy preliminary diagnoses, improving the efficiency of pet healthcare and helping both owners and veterinarians in the pursuit of better health outcomes for dogs.

Literature survey

Details of Paper- Title, author, conference /Journal	year	Methodology Used	Result	Limitations
Deep Learning-Based Pet Disease Diagnosis Using Skin and Fur Images A. Kumar, R. Sharma, S. Gupta	2023	Used CNN (ResNet50) for classification of pet skin diseases from image datasets.	Achieved 91% accuracy in detecting visible infections.	Focused only on images ignores symptom-based diagnostics.
Pet Care AI: A Smartphone-Based Health Monitoring Tool for Dogs S. Patel, R. Nair, P. Joshi	2021	Designed rule-based expert system for dog disease diagnosis based on user-input symptoms.	Assisted remote users to self-check dog health issues.	No machine learning or image processing; limited adaptability
PawSense: AI-IoT Enabled Smart Pet Care for Real-Time Health Monitoring Aravind G., Dr. Sasirekha S.P., Jeeva S., Selvaganesh D.	2025	Combined AI, IoT, and blockchain in a Flutter- based pet care system. Real-time health data from IoT collars processed with deep learning on cloud-edge architecture.	Enabled real-time anomaly detection and predictive alerts for pet health. Improved security and latency.	Only uses IoT sensor data; lacks integration of image and symptom-bas AI models.

Details of Paper-Title, author, conference /Journal	year	Methodology Used	Result	Limitations
Development of a Dog Health Score Using an Artificial Intelligence Disease Prediction Algorithm Based on Multifaceted Data S-C Kim & S. Kim	2024	Activity sensors (leash/wearables) monitoring behaviors (scratching, licking, swallowing, sleep) + AI algorithm to assign a “Health Score”.	Achieved ~87.5% concordance with veterinarian diagnoses.	Only behavioral data from sensor fusion.
A Multimodal Fusion Approach for Veterinary Disease Detection L. Chen, M. Zhao, Y. Li	2024	Combined CNN (images) and LSTM (text symptoms) for disease prediction in livestock.	Fusion improved accuracy by 12% over single-modal models	Model focused on livestock; not general systems..
Pet Pulse: Detecting Dog Diseases with TFLite, Booking Vet Consults” B. Abirami & T. Momithasree	2025	Mobile app combining symptom tracking + pulse monitoring + image processing (skin conditions) using TFLite on smartphone. I	Demonstrated ability for owners to detect pet skin/health issues via mobile app; improved accessibility	Limited described dataset; poor multimodal (symptoms + image)

Research Gap

While significant strides are being made in the field of veterinary technology and AI-based health analysis, several critical gaps persist that curtail the effectiveness of existing solutions for dog health monitoring. Most systems today work exclusively with either image-based data or text-based symptoms as inputs. Image-only diagnostic systems have gained much popularity, particularly for skin infections, rashes, or other external abnormalities. Many of the common ailments affecting dogs, such as digestive issues, viral infections, fever, or internal discomfort, do not necessarily show strong visual signs. Hence, image-only systems fail to capture the complete clinical features of such ailments and end up with incomplete or inaccurate assessments.

On the contrary, symptom-based tools rely solely on the descriptions given by the users, which introduces a high level of subjectivity. Pet owners can misinterpret symptoms, fail to notice important details, or give incomplete information about the symptoms exhibited. A symptom-only system, without visual confirmation, cannot distinguish between diseases that present similar symptoms. Most current text-based models also have high dependence on preliminary rule-based systems or small datasets; thus, generalizing across a wide array of breeds, ages, and health conditions is limited.

Another major research gap is that dog health monitoring lacks a multimodal dataset. Most publicly available datasets either comprise images of skin diseases or symptom descriptions, but rarely both together. This separation of symptom reports and images prevents AI models from learning the natural correlation between what a dog looks like and what it is experiencing. Without combined datasets, models cannot perform symptom-image cross-validation, which is necessary for accurate diagnosis. This gap also restricts the development of hybrid architectures using deep learning and machine learning together.

Moreover, much of the existing research completely ignores breed-specific variations. Small, medium, and large breeds of dogs differ considerably in their physiological, immunological, symptomatic, and disease manifestation characteristics. A condition that might appear acute in a small breed may be considered minor in a larger breed, and vice versa. Single-input systems, not really accounting for these subtleties, often provide generic or even misleading predictions. One of the major drawbacks with the current state of AI-powered veterinary tools is the lack of breed-aware models.

Besides these, accessibility and usability present further gaps. Many AI-powered diagnostic tools require some technical understanding from the users, high-quality images, or sets of symptoms that are too descriptive for an average pet owner to provide. This reduces effective adoption in practical real-life scenarios. The clear need is for a user-friendly, intuitive interface where a pet owner can simply upload an image, describe basic symptoms, and receive a reliable preliminary health assessment.

Finally, there is minimal work represented in the existing literature regarding the integration of deep learning-based visual analysis with machine learning/NLP-based symptom interpretation in the veterinary domain. While multimodal learning has been effective in human healthcare, it has not been extended to animal health monitoring in an effective way. This itself shows a significant unexplored frontier with strong potential for innovation.

OBJECTIVES :

1. To create an AI-based system that uses symptom data and images to identify and track health issues in dogs.
2. To use machine learning algorithms to study symptoms and deep learning to detect diseases from images.
- 3.To integrate (fuse) clinical and visual characteristics to predict diseases accurately and early.
4. To create a system that functions well with small, medium, and large dog breeds
5. To design an intuitive user interface that makes it simple for pet owners to enter symptoms and upload photos.
- 6.To support early diagnosis and preventive healthcare for veterinarians and pet owners.
7. To ensure our system can grow easily and can connect with new devices and new users by making health apps in the future

Overview of datasets

4.1 Description

Datasets are critical in enabling the multimodal learning approach, where both visual and textual information are put together to assess the health condition of a dog. Data was collected from publicly available sources, reliable data science communities, and resources such as Kaggle, Roboflow, and HuggingFace that host large datasets with great diversity and well-labeled examples, both of dog images and symptom descriptions.

These include photographs of dogs with visible health issues such as skin infections, rashes, fur loss, discoloration, wounds, and other visible abnormalities. These images are taken under different light conditions, angles, and environments that help in improving the robustness of the model. The datasets also contain healthy images to enable the model to identify the difference between normal and abnormal visual indicators.

Apart from the images, textual symptom datasets were obtained from veterinary symptom repositories and crowdsourced forums available in HuggingFace and similar platforms. Examples of symptoms described in such data include vomiting, diarrhea, loss of appetite, lethargy, coughing, itching, and abnormal color of waste. Most entries also contain extra contextual details like duration, frequency, and severity, which allows the model to understand a wide range of clinical presentations.

The combination of image and text-based datasets would facilitate the development of a multimodal system that can cross-verify symptoms by visual cues, enabling an even more accurate and reliable diagnosis. This dataset collection ensures that the model can evaluate both the external and internal indicators of the onset of an illness for early detection and to assist pet owners in making informed decisions.

4.2 Data Attributes / Features

Therefore, to construct a powerful multimodal AI framework, two important types of features are extracted from images and symptom descriptions. Such features enable the model to understand the health condition of the dog from two perspectives: visual and clinical.

A. Image Features

The key image-related attributes include:

Skin/fur coloration changes: Color patterns indicating possible infection, irritation, or inflammation in skin or fur.

Variations in skin texture: feeling coarse, dry with bumps or other irregularities.

Lesion characteristics Size, shape, and distribution of rashes, wounds, or patches

Alopecia: Circular or patchy areas of thinning or lost fur, usually due to allergic reactions or fungal infections.

Swelling or redness could be indicative of allergic reactions or bacterial infections.

Image metadata include resolution, lighting conditions, and background information.

Features are extracted using CNNs, which enable the model to learn the visual patterns associated with different disease categories.

B. Symptom Text Features

The text dataset contributes clinical attributes such as:

Primary symptoms: vomiting, diarrhea, itching, coughing, fatigue, less appetite, etc.

Behavioral changes: Increased restlessness, weakness, excessive licking, or social withdrawal.

Digestive indicators: Color of stool, its consistency, frequency, and patterns of abnormal waste.

Symptom duration: The length of time the symptom has been experienced.

Severity indicators include: frequency of episodes, intensity of discomfort, and effect on daily activity.

The features of text are extracted using NLP techniques like tokenization, embedding methods, and classification models.

C. Breed & Category Features

Since disease expression does vary across breeds, the dataset also includes:

Breed type: Labrador, German Shepherd, Beagle, Pug.

Breed size categories include small, medium, and large breeds.

Age and gender (if available): Useful for understanding susceptibility.

D. Label Features (Ground Truth)

Each entry in this dataset contains labels like:

Category of disease: dermatitis, mange, ear infection, malnutrition, allergy, etc.

Health status: healthy / mildly affected / severe

Symptom cluster mapping: Differential diagnosis based on symptoms

These labels enable supervised learning and allow the system to predict probable diseases.

CONCLUSION OF CAPSTONE PROJECT PHASE – 1

In phase-1 of our capstone project AI-Based Dog Health Monitoring System Using Dual-Input (Images + Symptom) , we successfully framed our problem statement. we identified how difficult it is for pet owners to recognize if their dogs is sick . during this phase, we researched existing methods like CNN for image and symptom-based classification models and multi model fusion techniques.

We got to know that combination of datasets is very rare, and initially collected datasets and whether to build app or website . we also checked practically possible

We got to know what we are suppose to do in phase 2 .

PLAN OF WORK FOR CAPSTONE PROJECT PHASE – 2

In Phase–2, our team will focus on identifying and finalizing suitable datasets for dog images and symptom information. After selecting the datasets, preprocessing steps such as cleaning, resizing, and encoding will be carried out. The development of the CNN model for images and the symptom analysis model will begin based on the available data. The team will begin basic testing of both models and check how they work together in the system. Our team is designing the system using a high-level modular approach that allows each component to work independently while still contributing to a unified prediction output. We are developing two major modules , an image analysis module and a symptom analysis module. For image processing, our team is implementing a CNN-based model that identifies visual issues such as skin infections, rashes . At the same time, we are building an NLP/ML model that interprets the symptoms entered by the pet owner, such as vomiting, appetite loss, or fatigue. Once both modules generate their predictions, we are integrating them through a multimodal fusion layer that combines visual and textual evidence to give a more accurate final diagnosis.